Comparative Optimization of the Rosenbrock Function

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Abstract

This report studies the generalized Rosenbrock objective through analytical derivatives, a nonlinear least-squares formulation, and seven iterative optimization configurations implemented directly in MATLAB. The experiments use the two-dimensional case with $b = 2$, Armijo backtracking with reduction factor 0.75, and the tolerances specified in the exercise. Numerical results are reported without treating stagnation or divergence as convergence.

1 Analytical properties

For $x \in \mathbb{R}^n$ and $b > 0$, the generalized Rosenbrock objective is

$$f(x) = \sum_{i=1}^{n-1} \left[b(x_{i+1} - x_i^2)^2 + (1 - x_i)^2 \right]. \tag{1}$$

The first and last components of its gradient are

$$g_1 = -4bx_1(x_2 - x_1^2) - 2(1 - x_1), \tag{2}$$

$$g_n = 2b(x_n - x_{n-1}^2), \tag{3}$$

while, for $1 < i < n$,

$$g_i = 2b(x_i - x_{i-1}^2) - 4bx_i(x_{i+1} - x_i^2) - 2(1 - x_i). \tag{4}$$

The Hessian is symmetric and tridiagonal. Its nonzero entries are

$$H_{1,1} = 2 - 4b(x_2 - 3x_1^2), \tag{5}$$

$$H_{i,i} = 2 + 2b - 4b(x_{i+1} - 3x_i^2), \qquad 1 < i < n, \tag{6}$$

$$H_{n,n} = 2b, \tag{7}$$

$$H_{i,i+1} = H_{i+1,i} = -4bx_i, \qquad 1 \le i < n. \tag{8}$$

Every term in (1) is nonnegative. The value zero is reached only if $1 - x_i = 0$ and $x_{i+1} - x_i^2 = 0$ for every $i < n$. Therefore the unique global minimizer is

$$x^\star = (1, \ldots, 1)^T, \qquad f(x^\star) = 0. \tag{9}$$

1.1 Nonlinear least-squares form

Define $r(x) \in \mathbb{R}^{2n-2}$ by

$$r_{2i-1}(x) = \sqrt{b}(x_{i+1} - x_i^2), \qquad r_{2i}(x) = 1 - x_i, qquad 1 \le i < n. \tag{10}$$

Then $f(x) = r(x)^T r(x)$. The Jacobian $J(x) \in \mathbb{R}^{(2n-2)\times n}$ is rectangular and sparse. For each $i = 1, \ldots, n-1$, its only nonzero entries in rows $2i - 1$ and $2i$ are

$$J_{2i-1,i} = -2\sqrt{b}\, x_i, qquad J_{2i-1,i+1} = \sqrt{b}, qquad J_{2i,i} = -1. \tag{11}$$

With the convention $f = r^T r$, the gradient identity is

$$\nabla f(x) = 2J(x)^T r(x). \tag{12}$$

Central finite differences were evaluated at two deterministic points for $n = 2, 3$, and 5. The largest relative discrepancy among the gradient, Hessian, Jacobian, and least-squares identity checks was below 1.7×10^{-10}. This verification does not require the Symbolic Math Toolbox.

2 Objective visualization

The experiments use $n = 2$ and $b = 2$. The objective was sampled with spacing 0.05 over $x_1 \in [-4, 4]$ and $x_2 \in [-5, 20]$.

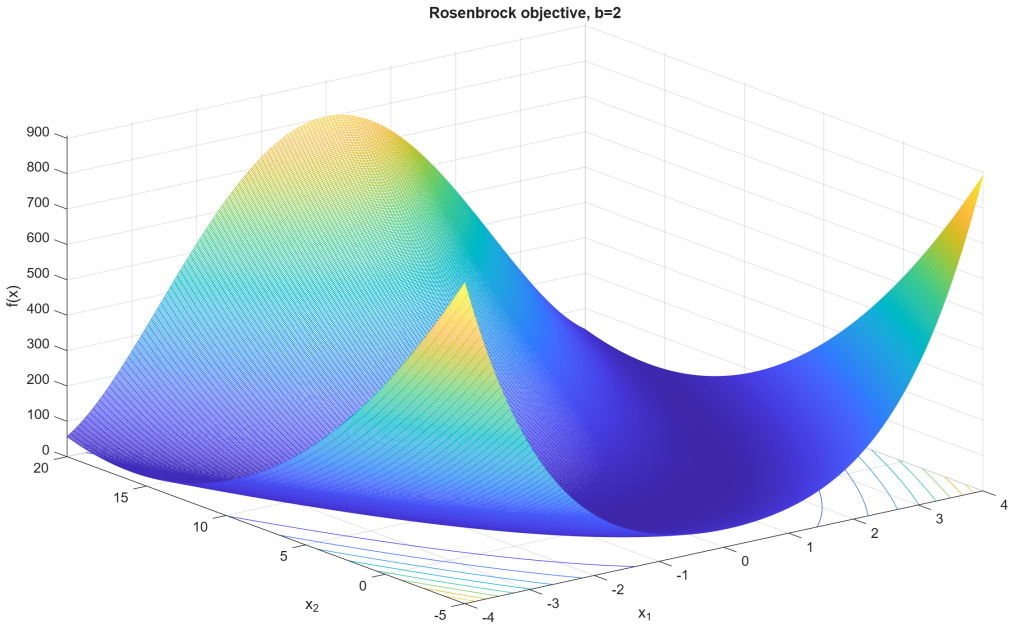


Figure 1: Surface and projected contours of the Rosenbrock objective for $b = 2$.

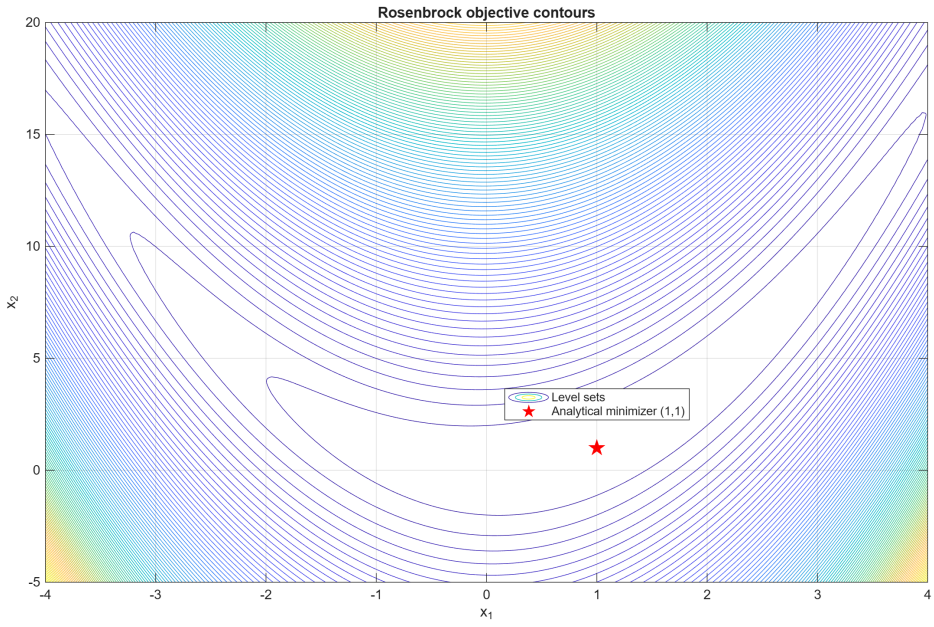


Figure 2: One hundred level sets and the analytical minimizer $(1, 1)^T$.

Figures 1 and 2 show the narrow curved valley that makes the problem sensitive to both the direction and the step length.

3 Fixed-step gradient descent

Starting from $x_0 = (0, 0)^T$, three fixed steps were tested with at most 1000 iterations. Relative changes are defined by

$$\delta_x = \frac{\|x_{k+1} - x_k\|_2}{\max(1, \|x_k\|_2)}, \qquad \delta_f = \frac{|f_{k+1} - f_k|}{\max(1, |f_k|)}. \tag{13}$$

Table 1: Fixed-step gradient descent from $x_0 = (0, 0)^T$.

Step	x_1	x_2	$f(x)$	$\|g\|_2$	Stop
1	-64.0000	16.0000	3.330×10^{7}	2.089×10^{6}	Divergence
10^{-2}	0.9924	0.9833	6.257×10^{-5}	6.842×10^{-3}	MaxIter
10^{-4}	0.1802	0.0041	6.737×10^{-1}	1.603	MaxIter

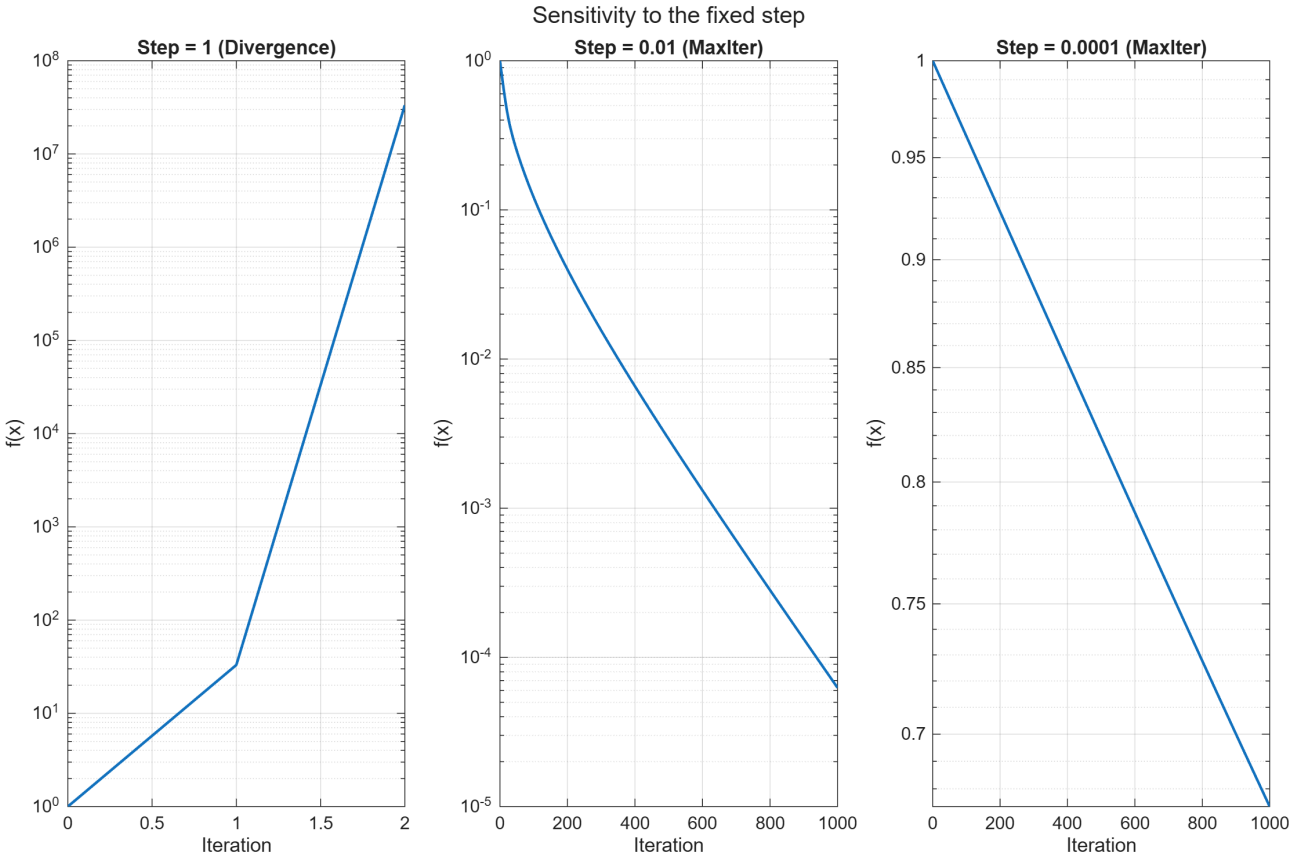


Figure 3: Objective histories for the three fixed steps. Divergence is retained rather than hidden.

The unit step is unstable from this starting point. The step 10^{-2} makes substantial progress but does not satisfy a stopping tolerance within the iteration limit. The smaller step 10^{-4} is stable but progresses too slowly. Figure 3 illustrates the sensitivity that motivates an adaptive line search.

4 Armijo backtracking

For a descent direction d_k, the accepted step satisfies

$$f(x_k + \alpha_k d_k) \le f(x_k) + c\alpha_k g_k^T d_k, \qquad c = 10^{-4}. \tag{14}$$

The search starts from $\alpha = 1$ and multiplies it by 0.75 after a failed trial. Stopping tests are applied only after a trial has been accepted.

From $(0, 0)^T$, gradient descent with Armijo reaches $x = (0.998729, 0.996562)^T$ after 139 iterations, with $f = 3.226 \times 10^{-6}$ and $\|g\|_2 = 5.858 \times 10^{-3}$. The stop is `TolF`, not `TolG`; it must therefore be interpreted as stagnation in the objective rather than full first-order convergence.

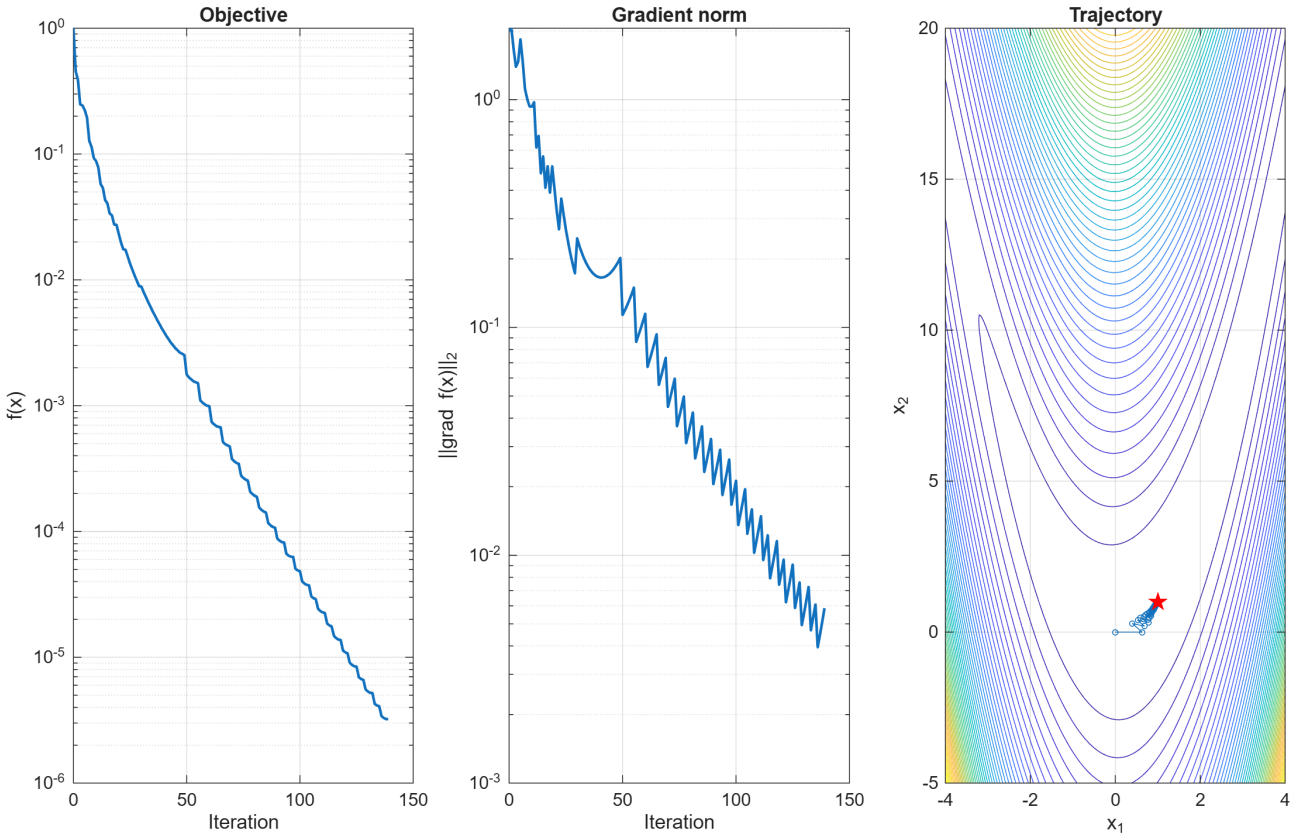


Figure 4: Objective, gradient norm, and trajectory for gradient descent with Armijo backtracking.

5 Comparison of optimization methods

The seven configurations are gradient descent with Armijo, nonlinear Fletcher–Reeves, Newton with a unit step, Newton with Armijo, inverse-BFGS, Gauss–Newton, and Levenberg–Marquardt. Every run reinitializes its internal state from the same starting vector. When the Newton Hessian is not positive definite, a diagonal spectral shift is added before solving the Newton system. BFGS is restarted with the identity matrix if its curvature condition fails.

Gauss–Newton solves a regularized version of $(J^T J)d = -J^T r$. Levenberg–Marquardt solves $(J^T J + \mu I)d = -J^T r$ and accepts a trial using the ratio between actual and predicted reduction. The damping parameter decreases when this ratio is at least 0.75 and increases when it is at most 0.25.

Table 2: Results from $x_0 = (-2, 10)^T$. Times are single-run measurements and are only indicative.

Method	$f(x)$	$\|g\|_2$	Iter.	Time (s)	Stop
Gradient Armijo	1.257×10^{-5}	1.128×10^{-2}	903	0.1314	TolF
Fletcher–Reeves	1.593×10^{-7}	1.226×10^{-3}	57	0.0917	TolF
Newton unit step	8.100×10^{1}	9.315×10^{1}	0	0.0987	Divergence
Newton Armijo	9.217×10^{-29}	5.939×10^{-14}	13	0.0138	TolG
BFGS	4.227×10^{-17}	5.618×10^{-9}	18	0.0195	TolG
Gauss–Newton	1.666×10^{-22}	1.217×10^{-11}	3	0.0050	TolG
Levenberg–Marquardt	8.089×10^{-14}	2.524×10^{-7}	5	0.0168	TolF

Table 3: Results from $x_0 = (4, -5)^T$. Times are single-run measurements and are only indicative.

Method	$f(x)$	$\|g\|_2$	Iter.	Time (s)	Stop
Gradient Armijo	2.181×10^{-6}	4.797×10^{-3}	149	0.0039	TolF
Fletcher–Reeves	2.601×10^{-8}	4.437×10^{-4}	74	0.0095	TolF
Newton unit step	6.174×10^{-23}	1.571×10^{-11}	5	0.0054	TolG
Newton Armijo	2.465×10^{-32}	9.930×10^{-16}	10	0.0012	TolG
BFGS	7.384×10^{-14}	1.343×10^{-6}	22	0.0054	TolF
Gauss–Newton	4.265×10^{-20}	1.947×10^{-10}	2	0.0059	TolG
Levenberg–Marquardt	1.313×10^{-17}	3.346×10^{-9}	4	0.0068	TolG

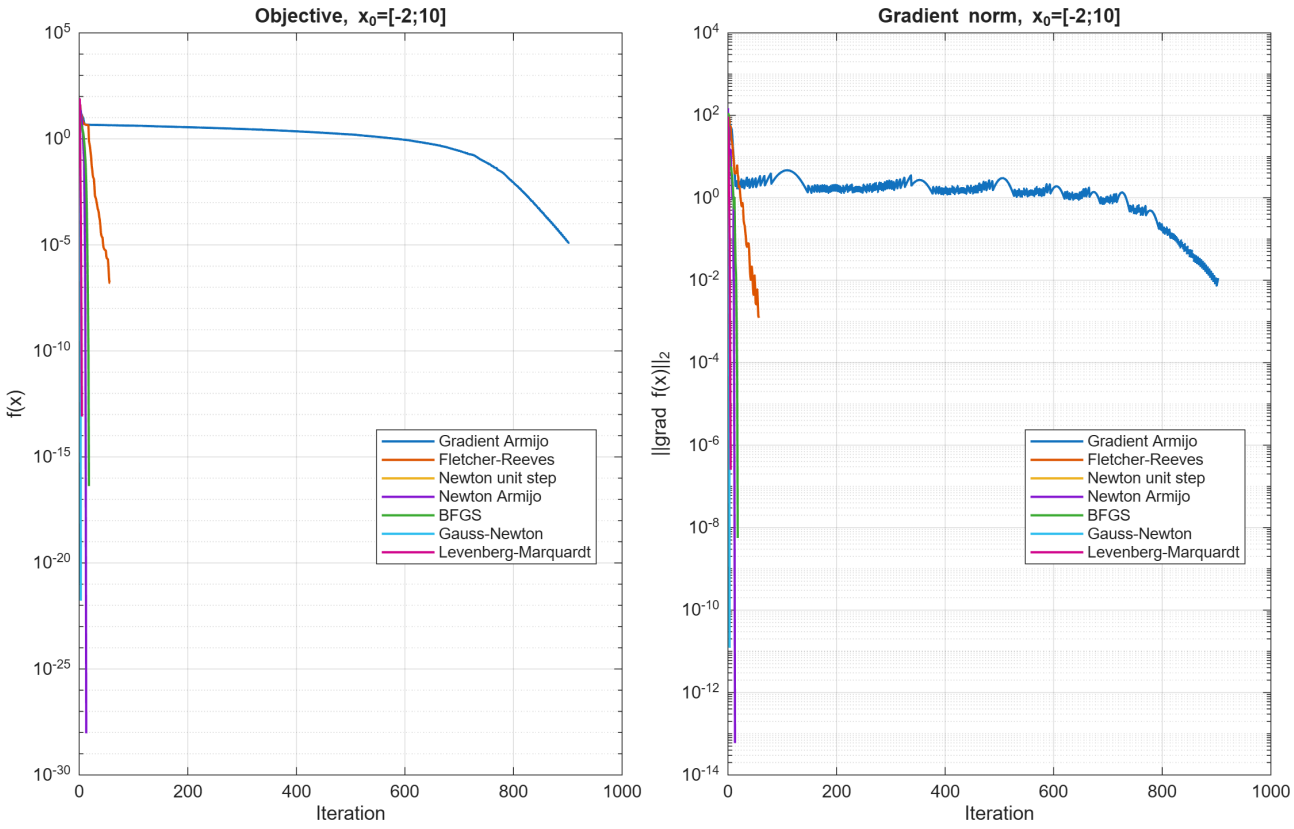


Figure 5: Objective and gradient histories from $x_0 = (-2, 10)^T$.

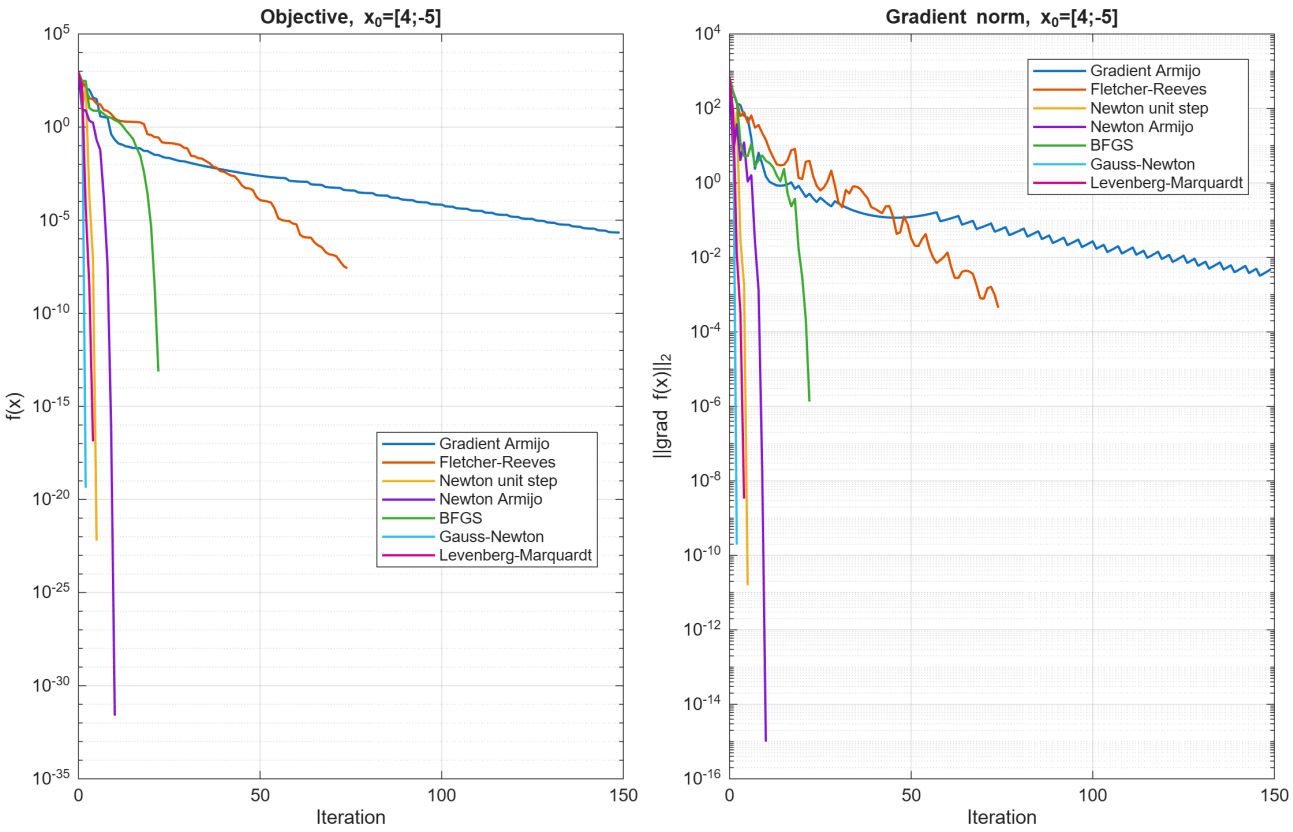


Figure 6: Objective and gradient histories from $x_0 = (4, -5)^T$.

The comparison in Figures 5 and 6 exposes a strong dependence on the starting point. The unit-step Newton method diverges immediately from $(-2, 10)^T$ but converges in five iterations from $(4, -5)^T$. Armijo globalization removes this failure in the first case. The nonlinear least-squares structure is especially favorable here: Gauss–Newton reaches the gradient tolerance in two or three accepted iterations, while the damped Levenberg–Marquardt method requires four or five. Gradient and Fletcher–Reeves runs that stop on `TolF` still have visibly larger gradient norms and should be described as stagnated rather than fully converged.

6 Conclusion

The Rosenbrock valley illustrates why a fixed step cannot simultaneously provide stability and fast progress over the full trajectory. Armijo backtracking improves robustness, but first-order methods still require many iterations. Curvature information substantially reduces the iteration count, provided that indefinite Hessians and failed trial steps are handled. For this specific objective, the residual representation gives Gauss–Newton and Levenberg–Marquardt a decisive advantage. These conclusions concern only the two-dimensional case with $b = 2$ and the two prescribed starting points; they are not a general ranking of optimization algorithms.

References

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